KUTAISI INTERNATIONAL UNIVERSITY

Scientific Computing

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#1.1

Classification of Ordinary Differential Equations

- Ordinary Differential Equations (ODEs)
- Classification by order
- Classification by type
- Classification by coefficients
- Classification by linearity
- Classification by variable type
- Q & A

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- **Riemann-Liouville Fractional Derivative:**
  - Let $n = \lceil \alpha \rceil$ (the smallest integer greater than or equal to α).
  - For $\alpha > 0$, the Riemann-Liouville fractional derivative of order α is:

$$ {}_{RL}D_t^{\alpha} f(t) = \frac{1}{\Gamma(n-\alpha)} \frac{d^n}{dt^n} \int_0^t \frac{f(\tau)}{(t-\tau)^{\alpha-n+1}} d\tau $$

  - For $0 < \alpha < 1$:

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  - For $\alpha > 0$, the Grünwald-Letnikov fractional derivative is:

$$ {}_{GL}D_t^{\alpha} f(t) = \lim_{h \to 0} \frac{1}{h^{\alpha}} \sum_{j=0}^{\lfloor t/h \rfloor} (-1)^j \binom{\alpha}{j} f(t - jh) $$

  - Where $\binom{\alpha}{j} = \frac{\Gamma(\alpha+1)}{\Gamma(j+1)\Gamma(\alpha-j+1)}$.

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- Key characteristics of linear ODEs:
  - The dependent variable y and its derivatives appear linearly (i.e., to the first power).
  - No products of y and its derivatives (e.g., yy' or $(y')^2$) are present.
  - No nonlinear functions of y or its derivatives (e.g., $\sin(y)$ or $e^{y'}$) are present.
- **Examples:**
  - $y'' + 2y' + y = \sin(x)$
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Classification of ODEs by Linearity: Linear ODEs

- **Linear ODEs:** An ODE is linear if it can be written in the form:

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  - Solution techniques often involve the characteristic equation and exponential functions.

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  - Solution methods often include power series, Frobenius method, and special functions.

Classification of ODEs by Variable Type: Dependent and Independent Variables

- **Dependent and Independent Variables:**
  - In an ODE, the **dependent variable** is the function we are trying to find (e.g., $y(x)$).
  - The **independent variable** is the variable with respect to which the derivatives are taken (e.g., x).
  - Example: In $\frac{dy}{dx} = f(x, y)$, y is the dependent variable and x is the independent variable.

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- **Autonomous ODEs:**
  - An ODE is autonomous if the independent variable does not explicitly appear in the equation.
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  - Example: $\frac{dy}{dt} = ay(1 - y)$.

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- **Systems of ODEs:**
  - A system of ODEs involves multiple dependent variables and their derivatives with respect to a single independent variable.
  - Example: $\frac{dx}{dt} = f(x, y)$, $\frac{dy}{dt} = g(x, y)$.

Classification of ODEs by Variable Type: Coupled/Decoupled Systems and Exact Equations

- **Coupled/Decoupled Systems:**
  - **Coupled systems:** The equations in the system are interdependent, meaning changes in one variable affect the others.
  - **Decoupled systems:** The equations can be solved independently.
  - Example (Coupled): $\frac{dx}{dt} = x + y$, $\frac{dy}{dt} = x - y$.
  - Example (Decoupled): $\frac{dx}{dt} = x$, $\frac{dy}{dt} = y$.

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- **Exact Equations:**
  - An ODE of the form $M(x, y)dx + N(x, y)dy = 0$ is exact if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.
  - Exact equations can be solved by finding a potential function $\psi(x, y)$ such that $\frac{\partial \psi}{\partial x} = M$ and $\frac{\partial \psi}{\partial y} = N$.
  - Example: $(2xy + \cos(x))dx + x^2 dy = 0$.

Classification of ODEs by Variable Type: Homogeneous/Nonhomogeneous (Revisited)

- **Homogeneous Equations (Revisited):**
  - In the context of first-order ODEs, a homogeneous equation is of the form $\frac{dy}{dx} = f(\frac{y}{x})$.
  - These equations can be solved using the substitution $v = \frac{y}{x}$.
  - Example: $\frac{dy}{dx} = \frac{x^2+y^2}{xy}$.

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- **Nonhomogeneous Equations (Revisited):**
  - An ODE that cannot be written in the homogeneous form is called nonhomogeneous.
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- **Linear Homogeneous/Nonhomogeneous (Revisited):**
  - Linear ODEs are classified as homogeneous if the function g(x) = 0, and nonhomogeneous if g(x) != 0.
  - This classification is independent of the homogeneous/nonhomogeneous classification of first order odes that rely on the f(y/x) structure.

Q & A